

Unit conversion to a system with $G = 1$ for EXP

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1 Definitions

Let the original (denoted as *old*) numerical gravitational constant be G_{old} . A particle in the old units has numeric mass m , position vector $\mathbf{r} = (x, y, z)$ and velocity vector $\mathbf{v} = (u, v, w)$. Define unit-scale factors as the number of old units contained in one new unit (denoted as *new*):

$$s_L \equiv \frac{L_{\text{new}}}{L_{\text{old}}}, \quad s_V \equiv \frac{V_{\text{new}}}{V_{\text{old}}}, \quad s_M \equiv \frac{M_{\text{new}}}{M_{\text{old}}}. \quad (1)$$

A numeric quantity in the new units (denoted with a prime) is obtained by dividing the old numeric value by the corresponding scale factor:

$$\mathbf{r}' = \frac{\mathbf{r}}{s_L}, \quad \mathbf{v}' = \frac{\mathbf{v}}{s_V}, \quad m' = \frac{m}{s_M}. \quad (2)$$

Time-scale factor:

$$s_T \equiv \frac{T_{\text{new}}}{T_{\text{old}}} = \frac{s_L}{s_V}.$$

2 Dimensional constraint for G and the core relation

Dimensional analysis gives $[G] = L V^2/M$. Thus numeric values transform as

$$G_{\text{new}} = G_{\text{old}} \frac{s_M}{s_L s_V^2}.$$

Requiring $G_{\text{new}} = 1$ gives the central constraint

$$\boxed{s_M = s_L s_V^2 / G_{\text{old}}}.$$

As a simple example, my usual recommendation to Gadget users is to keep their familiar length scale, velocity scale and time scale, and multiple the particle masses by G_{old} . Using these relations, we have $s_L = s_V = s_T = 1$ and $s_M = 1/G_{\text{old}}$ (see eq. 2).

3 Conversion formulas (apply to each particle)

Given $(m, \mathbf{r}, \mathbf{v})$ in old units:

$$\boxed{m' = \frac{m}{s_M}, \quad \mathbf{r}' = \frac{\mathbf{r}}{s_L}, \quad \mathbf{v}' = \frac{\mathbf{v}}{s_V}, \quad t' = \frac{t}{s_T} = \frac{t s_V}{s_L}}.$$

4 Solve-for formulas (pick any two scales, compute the third)

From $s_M = s_L s_V^2 / G_{\text{old}}$:

- Given s_L and s_V :

$$s_M = \frac{s_L s_V^2}{G_{\text{old}}}.$$

- Given s_L and s_M :

$$s_V = \sqrt{\frac{s_M G_{\text{old}}}{s_L}}.$$

- Given s_V and s_M :

$$s_L = \frac{s_M G_{\text{old}}}{s_V^2}.$$

5 Monopole (circular velocity) condition

Suppose an enclosed mass (in old units) is M_{enc} inside radius r_0 (old units). The old circular speed satisfies

$$v_{c,\text{old}}^2 = \frac{G_{\text{old}} M_{\text{enc}}}{r_0}.$$

In the new units (where $G_{\text{new}} = 1$), at the same physical radius (numerically $r'_0 = r_0/s_L$) the circular speed squared becomes

$$v_c'^2 = \frac{(M_{\text{enc}}/s_M)}{(r_0/s_L)} = \frac{M_{\text{enc}} s_L}{s_M r_0}.$$

Using $s_M = s_L s_V^2 / G_{\text{old}}$ this simplifies to

$$v_c'^2 = \frac{G_{\text{old}} M_{\text{enc}}}{s_V^2 r_0}$$

or equivalently

$$s_V = \sqrt{\frac{G_{\text{old}} M_{\text{enc}}}{v_c'^2 r_0}}.$$

Hence: if you demand a specific numeric circular speed value v'_c at physical radius r_0 , this uniquely determines s_V (independent of s_L). Once s_V is set, s_M follows from $s_M = s_L s_V^2 / G_{\text{old}}$; if you additionally set a target numeric radius r'_0 for r_0 , choose $s_L = r_0/r'_0$.

6 Summary table

Task	Formula
Core constraint	$s_M = \frac{s_L s_V^2}{G_{\text{old}}}$
Convert variables	$r' = r/s_L, \quad v' = v/s_V, \quad m' = m/s_M, \quad t' = t s_V/s_L$
Solve for s_M	$s_M = \frac{s_L s_V^2}{G_{\text{old}}}$
Solve for s_V	$s_V = \sqrt{\frac{G_{\text{old}} s_M}{s_L}}$
Solve for s_L	$s_L = \frac{G_{\text{old}} s_M}{s_V^2}$
Circular-velocity condition	$s_V = \sqrt{\frac{G_{\text{old}} M_{\text{enc}}}{v_c'^2 r_0}}$
Resulting s_M	$s_M = \frac{s_L s_V^2}{G_{\text{old}}} = s_L \frac{M_{\text{enc}}}{v_c'^2 r_0}$

7 Usage recipes

- Pick new length and velocity units directly: specify s_L and s_V , compute $s_M = s_L s_V^2 / G_{\text{old}}$, then rescale particle numbers with the conversion formulas.
- If you want $v_c' = v_*$ at radius r_0 for enclosed old mass M_{enc} : compute s_V from the circular-velocity formula, pick s_L (e.g. $s_L = r_0$ if you want $r_0' = 1$), then s_M follows.
- If you require a particular old mass M_* to become $m' = 1$: set $s_M = M_*$, then choose s_L (or s_V) and compute the remaining scale factor.